

COGS 45: Computational Cognitive Science

Lab Homework #4

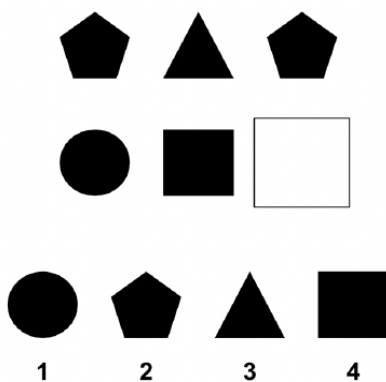
This week, we looked at how systems with **external memory** can improve **generalization ability** over Recurrent Neural Networks (RNNs). External memory models were inspired both by the architecture of the *in silico* systems we use every day (with distinct CPU and RAM), as well as of the high-level architecture mammalian brain (with distinct neocortical systems for slowly learning statistical structure and a hippocampal system for fast episodic updating in different circumstances as in McClelland et al. 1995).

First, please read the uploaded brief empirical Science paper by Marcus et al. (1999) on abstract rule-learning abilities in 7 month old infants¹. Marcus et al. use measurements of infant looking times to infer what the infant knows about the stimulus (i.e., *they attend longer to things that surprise them, so attending longer to rule-violations is evidence of knowing the rule*). This result was surprising at the time, because it showed that infants appear to either have an innate ability for rule-learning --- or, at least, were able to learn rules much more quickly than any neural network known to exist at the time could.

The “**identity-rules**” task from the Webb et al. (2021) reading re-creates the structure of the rule learning problem faced by infants in Marcus’s task, but uses *image-shapes* as input, rather than *syllables* in an artificial language. For example, rather than sequences of novel words (e.g., “*wo fe wo*”), as Marcus’s infants heard, the networks observe sequences of objects (*pentagon, triangle, pentagon*), that are new to the network, but in the same “ABA” pattern.

We attempt to re-create Webb et al.’s modeling results on the identity rules task here, and compare the performance of the ESNB to other models.

Example. Identity Task (answer is A).



¹ Also, here: <https://www.science.org/doi/epdf/10.1126/science.283.5398.77>

1. Train the networks on the minimal number of objects necessary to specify a task (5), and hold out 95 objects for testing (--m_holdout 95). Compare the results of an ESNB (--model_name ESNB) with the Neural Turing Machine (the Hybrid computing architecture we read about) (--model_name NTM) and Long-Short Term Memory (--model_name LSTM). (LSTMs are RNNs with trainable “gating” parameters for inputs/outputs and forgetting. We discussed them briefly in our first “Memory” lecture). These two other models can also store recent sequences of data to help them better process future information.

code is available in .ipynb uploaded to canvas

- a. Plot the **training loss over time** for all three models. Do so on the same graph for easy comparison. (as in Figure 4c in Webb et al.). Do the three models require the same amount of experience to achieve a given level of training loss?



Figure 1: Results for all three models (ESNB, Long-Short Term Memory, and Neural Turing Machine) with identity rules tasks (m=95) that represent training loss over time.

Response: The three models, ESNB, LTSM and NTM, do not require the same amount of experience to reach the same level of training loss. The ESNB comes to zero in about the 15th round of the training batch. While the NTM is improving over time reaching almost 43d training batch, the LSTM model is the one that stays longest with the training batch of 60.

This shows that the ESNB model has the strongest architectural inductive bias for the task, demonstrating it is good at separating the rule structure from the specific objects. On the other hand, the NTM and LSTM need to learn more indirectly from experience, requiring more time to train. As mentioned by Marcus et al. infants need

“a device that tracks statistical relationships and another that manipulates variables” than sequence memorization alone to be able to generalize abstract rules (Marcus et. al, 1999). Similarly, ESNB could be a better option to simulate infant learning.

- b. Plot a **bar graph** for each model’s performance on the task, when given **held-out TEST data (i.e., those objects the network has never seen in familiar relational configurations)** (as in Figure 3d in Webb et al.). How well does each model generalize the learned rule to new objects?

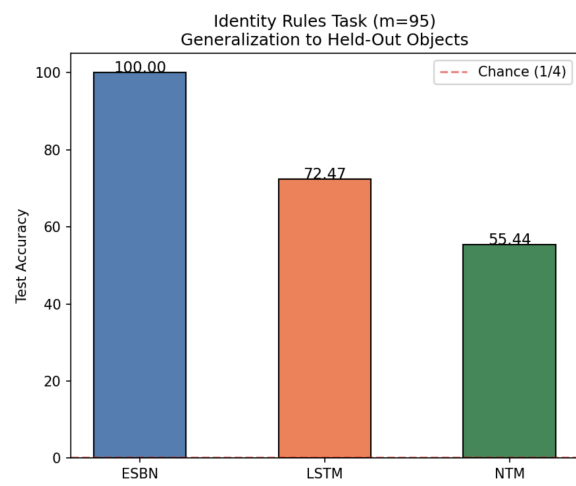


Figure 2: Results for all three models with identity rules tasks ($m=95$) that represent generalization to held-out objects. Results reflect test accuracy exceeding over 50% for each model.

Response: The bar graph demonstrates differences for all three models in how well each model is able to generalize the learned identity task to new held-out objects. The ESNB, the model that takes the shortest time to train, has 100% accuracy that it can learn the abstract rules and apply them to other objects it has never seen before. The LSTM generalizes somewhat well with about 72% accuracy, indicating that it has learned some sequences but cannot hold abstract information well. Finally, the NTM model falls short with 55% accuracy, failing to apply the identity rule task to new objects despite having an external memory. This shows that just having an external memory is not enough to generalize without having a structure and inductive bias.

- c & d. Do the same for `--m_holdout 75`. How does the gap between models change?



Figure 3: Results for all three models (ESBN, Long-Short Term Memory, and Neural Turing Machine) with identity rules tasks ($m=75$) that represent training loss over time.

Response: If we change the held-out objects from $m=95$ to $m=75$, the gap between the networks gets smaller. The graph shows that ESN is still faster than the LSTM and NTM, reaching 0 in the training batch 15. It demonstrates that the latter two models still require more training until they reach 0. Unlike $m=95$, the LSTM and NTM models show a better performance in this training because they have more training objects. Just what Web et. al purposes, they mainly acquire through statistical relationships and depend on examples.

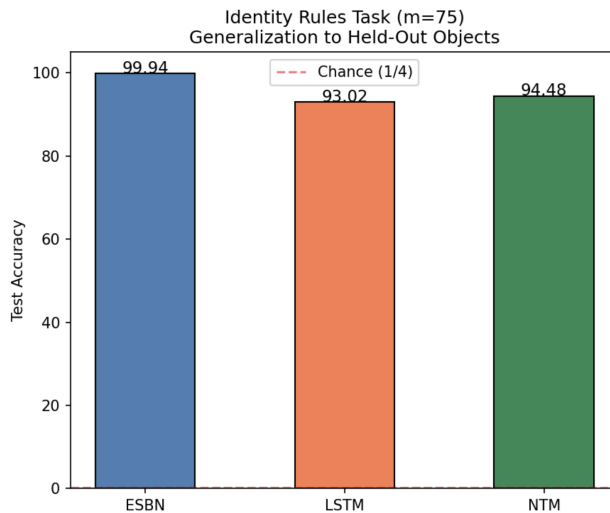


Figure 4: Results for all three models with identity rules tasks ($m=75$) that represent generalization to held-out objects. Results reflect test accuracy exceeding over 90% for each model.

Response: Compared to $m=95$, the gap between three models is smaller in $m=75$. All of the networks have a high accuracy rate exceeding 90% on generalizing to held-out

objects, with the ESN having highest accuracy of almost 100%, while the NTM reaching about 94% and the LSTM about 94%, respectively.

The bar shows that the LSTM and NTM models need more objects to be trained, demonstrating they are less similar to how infants learn that aligns with the empiricist views.

2. Marcus et al. claim that **statistical learning mechanisms** alone (like those in the RNN we used to predict the next character in Shakespeare's text, last week) are insufficient to explain the abstract rule-learning competency exhibited by 7 month old infants. Do your results suggest that augmenting those statistical learning mechanisms with an external memory enables the model to learn abstract rules capable of generalizing to novel inputs? Can *any* external memory system? If not, what seems to be required?

The statistical learning mechanisms alone are insufficient indeed because infants also need a "device" that can manipulate variables; however, having an external memory is not the same. For example, the LSTM fell short in $m=95$ because it required more objects for training that supports Marcus's argument that in rule generalization, statistical learning is not enough. The ESN, on the other hand, showed a high level of accuracy in fewer training batches with few objects, which demonstrated that the combination of external memory with structure and inductive bias is what can help generalize to novel objects. That is also true for the NTM model that despite having an external memory, it lacks necessary structures that the ESN model has.

3. Philosophers have long debated what aspects of the mind are innate and what is learned via experience. Briefly, those who think there are large bodies of innate knowledge are often called "rationalists" and those who emphasize the acquisition of knowledge via experience are often called "empiricists". How does your experiment here bear on that debate?

I personally think there should be a 'middle ground' where certain knowledge can be innate and certain others are learned in the later course of life. Learning abstract ideas as an infant but being able to apply them to novel concepts is deriving points from both rationalism and empiricism, I think.

I agree with David King, who argues that even comparing empiricism and rationalism in humans to neural networks is a wrong approach because (1) humans and these models have different underlying competencies, and (2) LLMs acquire through statistical grouping of data they're trained on. Thus, "it is questionable whether they have anything to tell us about the rationalist-empiricist debate at all" (King, 2024).

My experiment supports the argument that even though the neural networks show evidence for both rationalism and empiricism (the ‘middle ground’ I’m suggesting), they still process information much differently from humans that such a debate is irrelevant to pose in the first place.

References

1. King, D. (2024). Large language models and the rationalist empiricist debate. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2410.12895>
2. Marcus, G. F., Vijayan, S., Rao, S. B., & Vishton, P. M. (1999). Rule learning by Seven-Month-Old infants. *Science*, 283(5398), 77–80.
<https://doi.org/10.1126/science.283.5398.77>
3. Webb, T. W., Sinha, I., & Cohen, J. D. (2020, December 29). *Emergent Symbols through Binding in External Memory*. arXiv.org. <https://arxiv.org/abs/2012.14601>